

LUNAR BUILD MISSION PLAN — Test Site
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

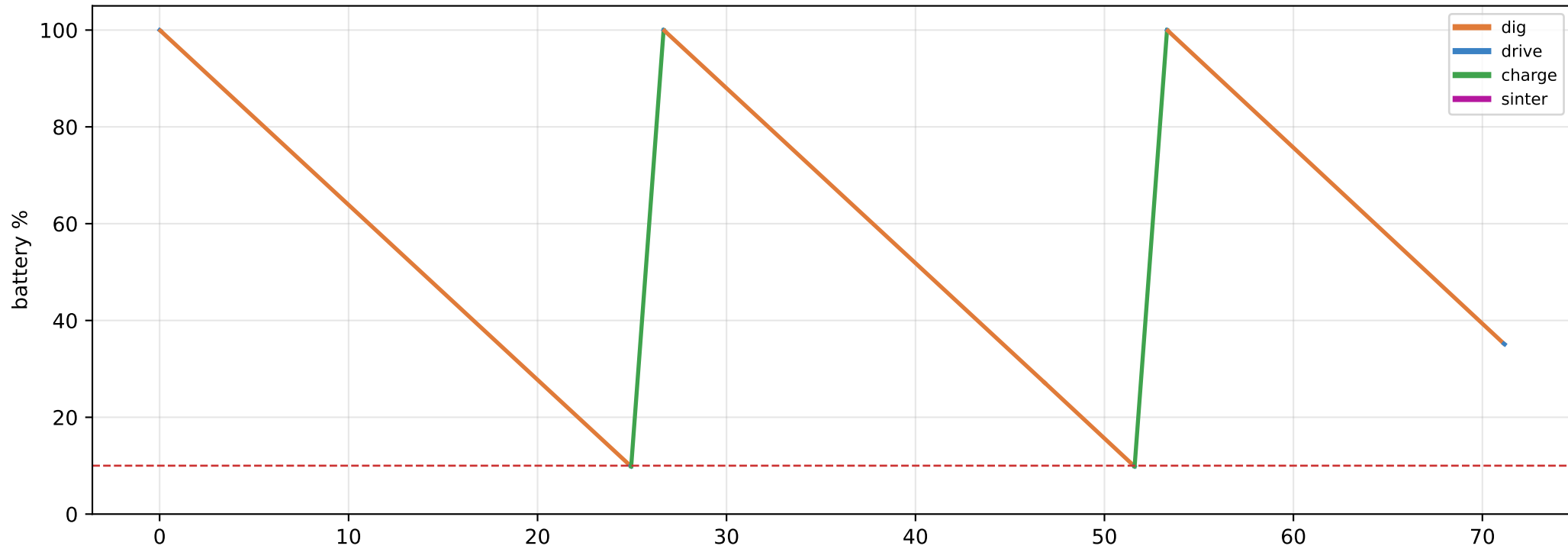
#	Trip	kind	Site (x,y)	Mass t	Duration	Energy (chg)
1	Level pad → Build berm	cutfill	(40,30)	1.82	46.8 h	1.6
2	Excavate spoil: Level pad	dig	(40,30)	0.94	24.3 h	0.8

MATERIAL BALANCE
cut 2.8 t → fill 1.8 t · surplus(spoil) 0.9 t · deficit(import) 0.0 t · sinter 0.00 t

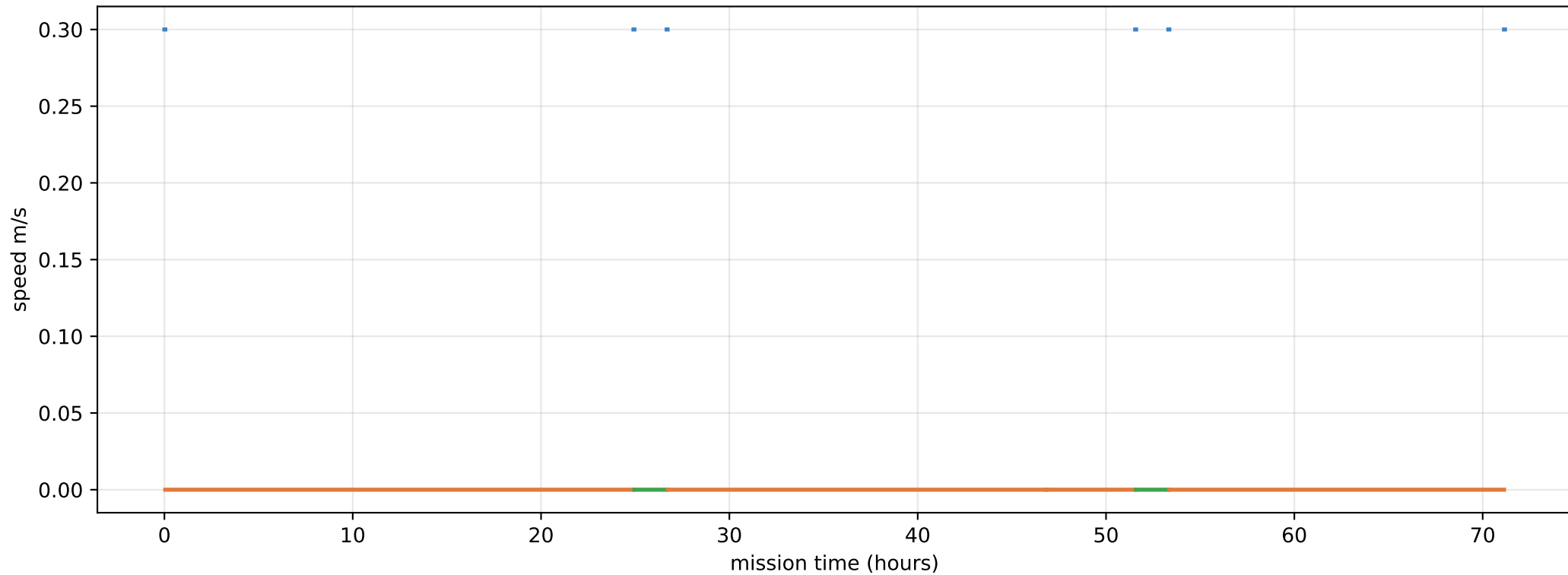
TOTALS project 3.0 d (71 h) moved 2.8 t (92 drum loads)
energy 11.8 MJ (2.5 charges, 2 recharge stops) drive 2.08 km
61 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell
per-sortie range 32 km (flat)
1 lunar day ~ 30 Earth-days (354 h light)
ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate

Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.

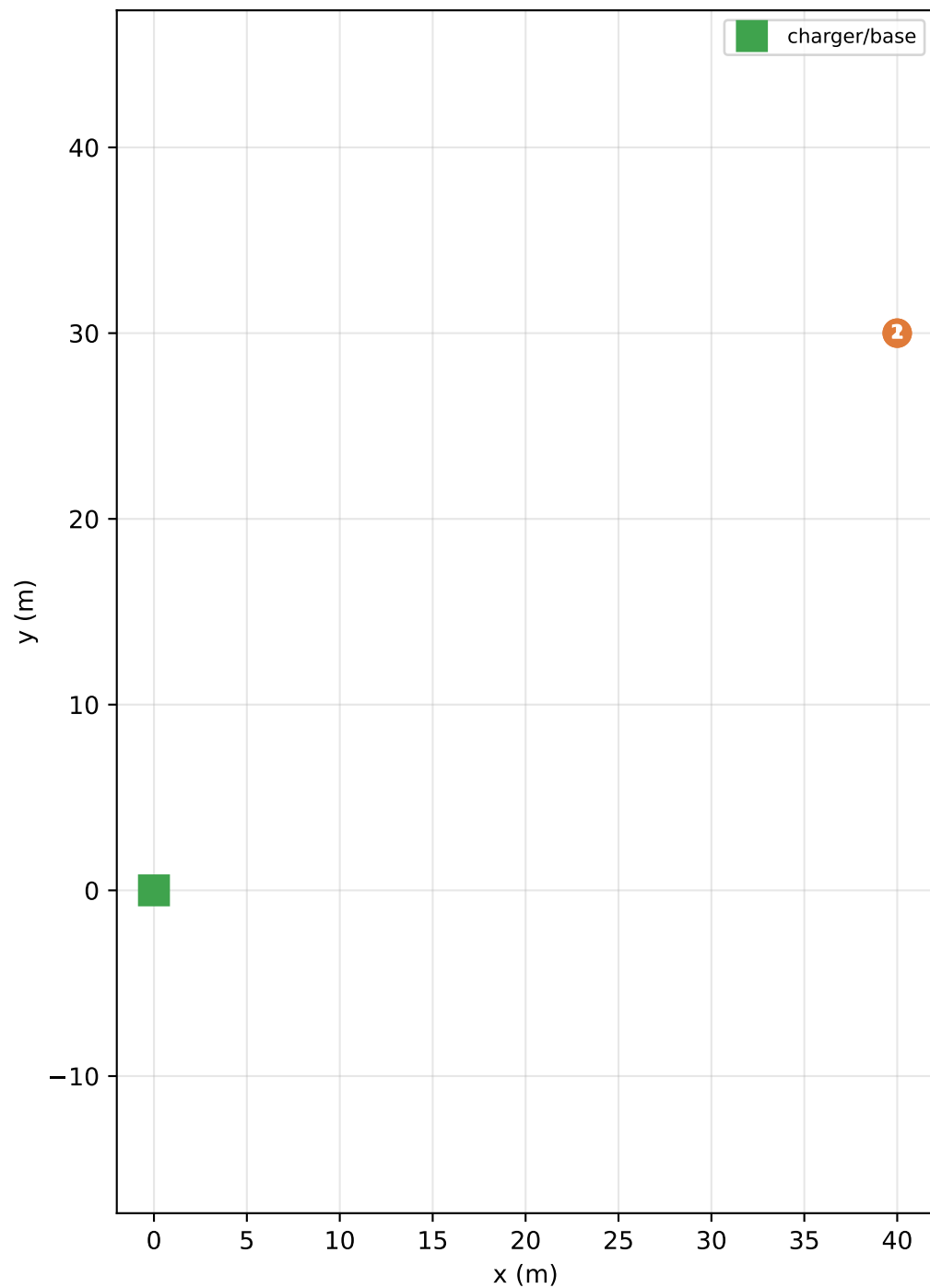
Battery draw over the planned project



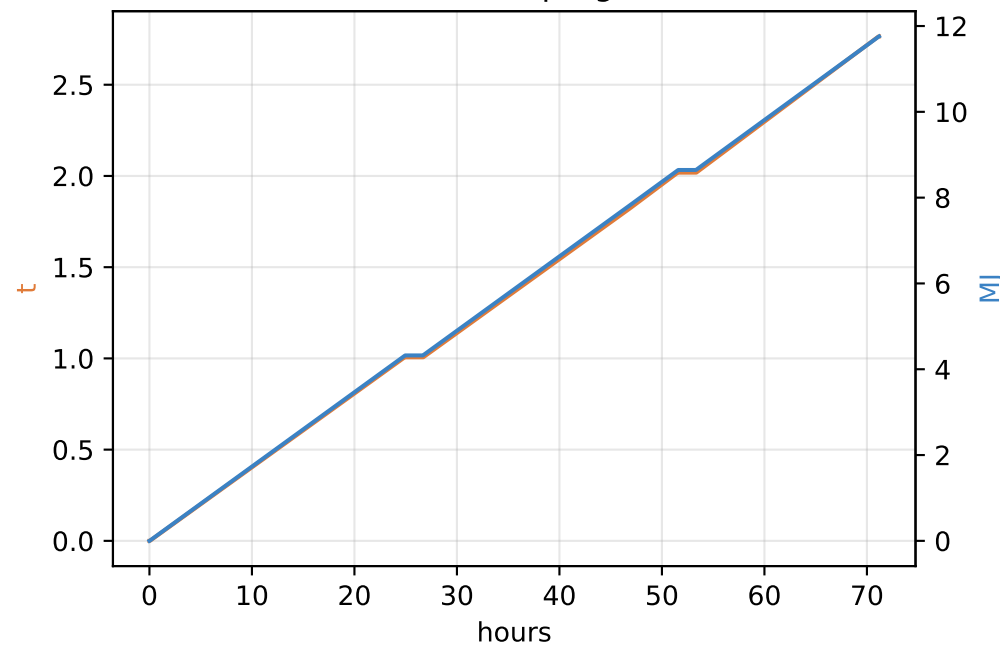
Speed profile



Site route + material flows (cut→fill)



Cumulative progress



Energy per trip (battery charges)

